

DataSpy Decision AI

Key Decisions Report

Generated by: DataSpy Technologies

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Executive Decision Summary

This report presents the recommended actions generated by the decision engine. Each item includes business rationale, priority, and supporting evidence.

Decision Overview Table

Rank	Priority	Decision Type	Action
	high	sales	
	high	sales	
	high	sales	
	medium	sales	

Detailed Recommendations

1. Not available

Priority: high

Decision Type: sales

Rank Order: Not available

Rationale: Top-performing products are currently a key revenue concentration area.

Supporting Evidence:

```
{
  "insight_reference": {
    "metric_type": "revenue",
    "metric_column": "total_amount",
    "top_products": [
      "Harrier",
      "Nexon",
      "Tiago",
      "Altroz",
      "Safari"
    ],
    "top_5_total_value": 849899640,
    "top_5_share_percent": 100,
    "product_count": 5
  },
}
```

```

"data_reference": {
  "product_column": "product",
  "value_column": "total_amount",
  "dataset_quality_label": "high",
  "dataset_quality_score": 0.9825
},
"schema_reference": {
  "product_role_confidence": 1,
  "value_role_confidence": 1
},
"capability_reference": {
  "sales": {
    "status": "enabled",
    "confidence": 0.6,
    "enable_reason": "minimum role coverage achieved; required core roles present",
    "role_breakdown": {}
  }
}
}

```

2. Not available

Priority: high

Decision Type: sales

Rank Order: Not available

Rationale: High dependence on a limited product set raises resilience concerns.

Supporting Evidence:

```

{
  "insight_reference": {
    "metric_type": "revenue",
    "metric_column": "total_amount",
    "top_products": [
      "Harrier",
      "Nexon",
      "Tiago",
      "Altroz",
      "Safari"
    ],
    "top_5_total_value": 849899640,
    "top_5_share_percent": 100,
    "product_count": 5
  },
  "data_reference": {
    "product_column": "product",
    "value_column": "total_amount",

```

```

"dataset_quality_label": "high",
"dataset_quality_score": 0.9825
},
"schema_reference": {
"product_role_confidence": 1,
"value_role_confidence": 1
},
"capability_reference": {
"sales": {
"status": "enabled",
"confidence": 0.6,
"enable_reason": "minimum role coverage achieved; required core roles present",
"role_breakdown": {}
}
}
}

```

3. Not available

Priority: high

Decision Type: sales

Rank Order: Not available

Rationale: This helps prevent over-commitment when projected demand signals weaken.

Supporting Evidence:

```

{
"forecast_reference": {
"forecast_mode": "predictive_forecast",
"model_used": "seasonal_naive_7",
"trend": "decreasing",
"growth_percent": -37.65,
"reliability_label": "Medium",
"reliability_score": 0.65,
"decision_usability_flag": "usable_with_caution",
"validation_metrics": {
"mae": 2677656.4,
"rmse": 3153881.9893,
"mape": 57.0363,
"smape": 35.5693,
"wape": 31.9921
},
"warnings": [
"Duplicate timestamp ratio is 84.83%, which exceeds allowed threshold 10.00%",
"Target series shows moderate volatility.",
"Validation error is high."
],

```

```
"candidate_diagnostics": [  
  {  
    "model_name": "seasonal_naive_7",  
    "mae": 2677656.4,  
    "is_stable": true,  
    "notes": [  
      "stability_score=0.716",  
      "consistency_score=0.0"  
    ]  
  },  
  {  
    "model_name": "drift_trend_model",  
    "mae": 2913621.8867,  
    "is_stable": true,  
    "notes": [  
      "stability_score=0.829",  
      "consistency_score=0.0"  
    ]  
  },  
  {  
    "model_name": "seasonal_weekday_average",  
    "mae": 3437581.6611,  
    "is_stable": true,  
    "notes": [  
      "stability_score=0.816",  
      "consistency_score=0.0"  
    ]  
  },  
  {  
    "model_name": "naive_last_value",  
    "mae": 4454032.2,  
    "is_stable": false,  
    "notes": [  
      "High variation across folds detected",  
      "stability_score=0.615",  
      "consistency_score=0.0"  
    ]  
  },  
  {  
    "model_name": "xgboost_lag_model",  
    "mae": null,  
    "is_stable": false,  
    "notes": [  
      "validation_failed=Feature frame is empty after feature engineering."  
    ]  
  }  
]
```

```

},
"data_reference": {
  "dataset_quality_label": "high",
  "dataset_quality_score": 0.9825,
  "missing_percent_overall": 0,
  "duplicate_rows": 0
},
"capability_reference": {
  "sales": {
    "status": "enabled",
    "confidence": 0.6,
    "enable_reason": "minimum role coverage achieved; required core roles present",
    "role_breakdown": {}
  },
  "inventory": {
    "status": "not_enabled",
    "confidence": 0.6,
    "enable_reason": "missing required roles: stock",
    "role_breakdown": {}
  }
}
}
}

```

4. Not available

Priority: medium

Decision Type: sales

Rank Order: Not available

Rationale: Forecast signal exists, but reliability conditions still limit how strongly it should be used.

Supporting Evidence:

```

{
  "forecast_reference": {
    "forecast_mode": "predictive_forecast",
    "model_used": "seasonal_naive_7",
    "trend": "decreasing",
    "growth_percent": -37.65,
    "reliability_label": "Medium",
    "reliability_score": 0.65,
    "decision_usability_flag": "usable_with_caution",
    "validation_metrics": {
      "mae": 2677656.4,
      "rmse": 3153881.9893,
      "mape": 57.0363,
      "smape": 35.5693,
      "wape": 31.9921
    }
  }
}

```

```
,
"warnings": [
  "Duplicate timestamp ratio is 84.83%, which exceeds allowed threshold 10.00%.",
  "Target series shows moderate volatility.",
  "Validation error is high."
],
"candidate_diagnostics": [
  {
    "model_name": "seasonal_naive_7",
    "mae": 2677656.4,
    "is_stable": true,
    "notes": [
      "stability_score=0.716",
      "consistency_score=0.0"
    ]
  },
  {
    "model_name": "drift_trend_model",
    "mae": 2913621.8867,
    "is_stable": true,
    "notes": [
      "stability_score=0.829",
      "consistency_score=0.0"
    ]
  },
  {
    "model_name": "seasonal_weekday_average",
    "mae": 3437581.6611,
    "is_stable": true,
    "notes": [
      "stability_score=0.816",
      "consistency_score=0.0"
    ]
  },
  {
    "model_name": "naive_last_value",
    "mae": 4454032.2,
    "is_stable": false,
    "notes": [
      "High variation across folds detected",
      "stability_score=0.615",
      "consistency_score=0.0"
    ]
  },
  {
    "model_name": "xgboost_lag_model",
    "mae": null,
```

```
"is_stable": false,
"notes": [
  "validation_failed=Feature frame is empty after feature engineering."
]
},
"data_reference": {
  "dataset_quality_label": "high",
  "dataset_quality_score": 0.9825,
  "missing_percent_overall": 0,
  "duplicate_rows": 0
},
"capability_reference": {
  "sales": {
    "status": "enabled",
    "confidence": 0.6,
    "enable_reason": "minimum role coverage achieved; required core roles present",
    "role_breakdown": {}
  },
  "inventory": {
    "status": "not_enabled",
    "confidence": 0.6,
    "enable_reason": "missing required roles: stock",
    "role_breakdown": {}
  }
}
```